Discussion Section 1
STAT 132 , Winter 2025

1. Suppose that the prior distribution of some parameter θ is a beta distribution for which the mean is $1/3$ and the variance is $1/45$. Determine the prior pdf of θ.

2. Suppose that the time a student spends studying each week follows an exponential distribution with rate parameter λ. After randomly sampling 3 students, you found that they studied 2, 2.5, and 3 hours respectively. Write out the likelihood function for λ, and then draw a sketch of this function.

3. Suppose that the time a student spends studying each week follows an exponential distribution with rate parameter λ. Your friend suggests a uniform distribution over the interval $[0, 10]$ for λ. Is this a reasonable prior distribution for this problem? Why or why not?

4. Suppose that the proportion p of defective items in a large manufactured lot is known to be either 0.1 or 0.2. Come up with a reasonable prior distribution for this problem, and then write out $\pi(p)$.

5. Suppose that the proportion p of defective items in a large manufactured lot is unknown. The following prior distribution is used:

$$\pi(p) = \begin{cases} 2(1-p), & \text{for } 0 \leq p \leq 1 \\ 0, & \text{otherwise} \end{cases} .$$

   What is the prior mean and variance for p?